

Q and A: A New Standard of Care for Germline *BRCA1* and/or *BRCA2* Mutation Carriers With Early-Stage Breast Cancer

Neelam V. Desai, MD¹; Dana Zakalik, MD²; Mark R. Somerfield, PhD³; and Nadine M. Tung, MD¹

Approximately 3%-5% of patients with breast cancer have an inherited (ie, germline) pathogenic or likely pathogenic variant (mutation) in *BRCA1* or *BRCA2*.^{1,2} Until now, systemic therapy for patients with early-stage breast cancer has not differed on the basis of the presence of an underlying germline *BRCA1* and/or *BRCA2* mutation (g*BRCA1*m and/or g*BRCA2*m). However, recently, the OlympiA trial demonstrated a significant improvement in 3-year invasive disease-free survival (iDFS); 85.9% v 77.1%, hazard ratio 0.58, $P < .001$ and distant disease-free survival (87.5% v 80.4%, $P < .001$) with the addition of 1 year of adjuvant olaparib, a poly (adenosine diphosphate-ribose) polymerase (PARP) inhibitor, to standard therapy in g*BRCA1*m and/or g*BRCA2*m carriers with high-risk early-stage human epidermal growth factor receptor 2 (HER2)-negative breast cancer.³ These practice-changing results signal a major advancement in the treatment of early-stage *BRCA*-associated breast cancer, and US Food and Drug Administration (FDA) approval of olaparib is expected.

To offer a speedy response to the OlympiA trial analyses, ASCO recently published a rapid recommendation update of its hereditary breast cancer guideline to address the role of PARP inhibitors in patients with early-stage, HER2-negative breast cancer and germline *BRCA* mutations.⁴ This companion Q and A article addresses some of the questions that clinicians will face as they implement results of the OlympiA trial and the rapid recommendation update into clinical practice.

Question: What patient population should be considered for adjuvant olaparib for 1 year on the basis of the OlympiA trial?

The OlympiA trial included patients with early-stage, HER2-negative breast cancer at high risk for recurrence, who harbored germline *BRCA1* and/or *BRCA2* mutations; patients with a somatic or tumor-identified mutation alone were not eligible for this trial. For patients who had surgery first (adjuvant), eligibility included either triple-negative breast cancer (TNBC) with tumors at least 2 cm or nodal involvement or

hormone receptor (HR)-positive cancers with at least four involved nodes. For patients who received chemotherapy first (neoadjuvant), eligibility required TNBC with *any* residual disease at surgery or HR-positive breast cancer with residual disease plus a pretreatment Clinical post-treatment Pathologic Stage as well as Estrogen receptor status and tumor Grade (CPS+EG)⁵ score ≥ 3 . The benefit of olaparib for patients who fall outside of these eligibility criteria is unknown. The pretreatment clinical stage and post-treatment pathologic stage as well as estrogen receptor status and tumor grade score can be calculated using the MD Anderson online calculator⁶ (Table 1).

Question: Should pembrolizumab be used for g*BRCA1*m and/or g*BRCA2*m carriers with TNBC? If so, how should it be incorporated with olaparib?

On July 26, 2021, the FDA approved the addition of pembrolizumab to anthracycline plus taxane and carboplatin-based neoadjuvant chemotherapy for the treatment of stage II-III TNBC on the basis of the KEYNOTE-522 trial,⁷ which showed an improvement in both pathologic complete response rate and event-free survival. This trial did not provide information about germline *BRCA* status, but approximately 15% of all patients with TNBC harbor a g*BRCA1*m and/or g*BRCA2*m.⁸ Since it was not the standard of care when OlympiA was written, patients in OlympiA did not receive pembrolizumab.

Question: Should neoadjuvant pembrolizumab be used in g*BRCA1*m and/or g*BRCA2*m carriers with TNBC who meet eligibility for Keynote 522 (ie, tumor size ≥ 2 cm or nodal involvement)?

Unfortunately, results of the KEYNOTE-522 trial by *BRCA* status are not available. It is known that the rate of pathologic complete response with chemotherapy is higher in g*BRCA1*m and/or g*BRCA2*m carriers with TNBC than in nonmutation carriers.^{9,10} However, given the significant risk of relapse associated with stage II-III TNBC, it is reasonable to use pembrolizumab in the neoadjuvant setting for g*BRCA1*m and/or g*BRCA2*m carriers with TNBC who meet criteria for KEYNOTE-522. In the absence of specific data, clinical judgment

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TABLE 1. Indications for 1 Year of Adjuvant Olaparib Therapy **gBRCA1m and/or gBRCA2m (not just tumor-identified or somatic mutation alone)**

HER2-negative early-stage breast cancer as follows
Surgery first (adjuvant)
TNBC: node-positive or tumor ≥ 2 cm
HR-positive: ≥ 4 positive nodes
Chemotherapy first (neoadjuvant)
TNBC: residual disease at surgery
HR-positive: residual disease and a CPS + EG score ≥ 3

Abbreviations: CPS, the pretreatment clinical and post-treatment pathologic stage; EG, estrogen receptor status and tumor grade; gBRCA1m and/or gBRCA2m, germline *BRCA1* and/or *BRCA2* mutation; HER2, human epidermal growth factor receptor 2; TNBC, triple-negative breast cancer.

is required and factors such as baseline tumor burden may be considered.

Question: Should pembrolizumab be added to olaparib for gBRCA1m and/or gBRCA2m carriers with TNBC who have residual disease after neoadjuvant chemotherapy?

Given the results of OlympiA, these patients should receive 1 year of olaparib after chemotherapy, surgery, and radiation are completed. There are no data to address whether the addition of pembrolizumab in this setting would improve outcomes. There are safety data for the combination of olaparib and pembrolizumab in several other cancers (eg, ClinicalTrials.gov identifier: [NCT02861573](https://clinicaltrials.gov/ct2/show/study/NCT02861573)). Thus, for gBRCA1m and/or gBRCA2m carriers with TNBC who receive chemotherapy and pembrolizumab in the neoadjuvant setting, it is reasonable to consider combining olaparib with pembrolizumab for those with residual disease, recognizing that the benefit for the combination is unknown.

Question: For gBRCA1m and/or gBRCA2m patients with TNBC and residual disease after neoadjuvant chemotherapy, should the preferred agent be olaparib or capecitabine?

On the basis of the CREATE-X trial,¹¹ patients with TNBC and residual disease after neoadjuvant chemotherapy are currently treated with adjuvant capecitabine chemotherapy. Although results for CREATE-X according to gBRCA1m and/or gBRCA2m status are not available, it is presumed that approximately 15% of patients with TNBC in this trial had an underlying gBRCA1m and/or gBRCA2m.⁸ Adjuvant capecitabine was not allowed in the OlympiA trial since it is not yet the standard of care. There are no data directly comparing the efficacy of adjuvant olaparib and capecitabine in gBRCA1m and/or gBRCA2m carriers with early-

stage TNBC. However, in the metastatic setting, olaparib improved progression-free survival in gBRCA1m and/or gBRCA2m carriers when compared with standard chemotherapy in the OlympiAD trial and nearly half of those treated with chemotherapy received capecitabine.¹² Thus, in gBRCA1m and/or gBRCA2m carriers with early-stage TNBC who have residual disease after neoadjuvant chemotherapy, olaparib appears to be the preferred agent, although data comparing olaparib and capecitabine in the early-stage setting are lacking.

Question: For gBRCA1m and/or gBRCA2m patients with HR-positive, HER2-negative, early-stage, and high-risk breast cancer, should adjuvant olaparib or a cyclin-dependent kinase (CDK) 4/6 inhibitor be used?

On October 12, 2021, the FDA approved abemaciclib along with endocrine therapy for adjuvant treatment of patients with HR-positive, HER2-negative, node-positive, early-stage breast cancer at high risk of recurrence and a Ki-67 score $\geq 20\%$.¹³ This was based on the monarchE trial,¹⁴ which showed a 3-year iDFS of 88.8% vs 83.4% (hazard ratio, 0.7, $P < .0001$) favoring the addition of two years of adjuvant abemaciclib over standard endocrine therapy alone. However, the PALLAS¹⁵ and PENELOPE-B¹⁶ trials failed to show similar benefit with adjuvant palbociclib, although differences in patient selection, follow-up time, and CDK 4/6 inhibitors make study comparisons challenging. Data on the gBRCA1m and/or gBRCA2m status of the patients enrolled in the monarchE trial are not available. The criteria for olaparib use in gBRCA1m and/or gBRCA2m carriers with HR-positive disease were more restrictive in OlympiA, requiring ≥ 4 involved nodes or significant tumor burden after neoadjuvant chemotherapy. Although cross-trial comparison is problematic, the 3-year iDFS benefit and hazard ratios appear similar in the OlympiA and monarchE trials. For gBRCA1m and/or gBRCA2m carriers who have surgery first and have 1-3 involved nodes, abemaciclib can be considered. For those with ≥ 4 involved nodes, either olaparib or abemaciclib is an option with differing toxicity profiles, which may guide treatment selection. There are no data combining a CDK 4/6 inhibitor and a PARP inhibitor, although trials are ongoing (ClinicalTrials.gov identifier: [NCT03685331](https://clinicaltrials.gov/ct2/show/study/NCT03685331)).

In summary, for patients with gBRCA1m and/or gBRCA2m and high-risk early stage HER2-negative breast cancer who meet the OlympiA trial eligibility, addition of adjuvant olaparib for 1 year is now the new standard of care. Further research is needed to better define optimal ways to integrate various targeted therapies to improve patient outcomes.

AFFILIATIONS

¹Division of Hematology-Oncology, Department of Medicine, Beth Israel Deaconess Medical Center, and Harvard Medical School, Boston, MA

²Beaumont Health, Royal Oak, MI

³American Society of Clinical Oncology, Alexandria, VA

CORRESPONDING AUTHOR

Neelam V. Desai, MD, Beth Israel Deaconess Medical Center,
330 Brookline Ave, Shapiro 9—Breast Oncology, Boston, MA 02215;
e-mail: ndesai@bidmc.harvard.edu.

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AUTHOR CONTRIBUTIONS

Conception and design: All authors

Administrative support: Mark R. Somerfield

Collection and assembly of data: Dana Zakalik, Mark R. Somerfield

Data analysis and interpretation: Neelam V. Desai, Dana Zakalik, Mark R. Somerfield

Manuscript writing: All authors

Final approval of manuscript: All authors

Accountable for all aspects of the work: All authors

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST**Q and A: A New Standard of Care for Germline *BRCA1* and/or *BRCA2* Mutation Carriers With Early-Stage Breast Cancer**

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Nadine M. Tung

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